HDOC Day-75

13-Nov-2026 | Enumeration in C

Question 1: Explicit Values in an Enumeration

Problem Statement: Assign explicit values starting from 10 in an enumeration and print them. Input: Nil. Output: A=10, B=11, C=12.

Algorithm / Logic

1. Define an enumeration with A explicitly initialized to 10.

2. Because enumeration values increase by 1 automatically, B becomes 11 and C becomes 12.

3. Print the integer values of A, B and C using printf().

C Program

```
#include <stdio.h>

enum Values {
    A = 10,
    B,
    C
};

int main(void) {
    printf("A=%d, B=%d, C=%d\n", A, B, C);
    return 0;
}
```

Explanation

In C, an enum assigns integer constants to names. When the first member A is explicitly assigned 10, the following members receive consecutive values automatically. Therefore, B is 11 and C is 12.

Test Cases

Test Case	Input	Expected Output
1	Nil	A=10, B=11, C=12

Sample Output

```
A=10, B=11, C=12
```

Result

The program successfully assigns enumeration values starting from 10 and prints A=10, B=11 and C=12.

Question 2: Menu Operations Using Enumeration and Switch

Problem Statement: Use enumeration to represent menu choices (ADD, SUBTRACT, MULTIPLY) and perform the selected arithmetic operation using switch. Example Input: ADD 10 20. Output: 30.

Algorithm / Logic

1. Define an enumeration for ADD, SUBTRACT and MULTIPLY.

2. Read the operation as a string followed by two integers.

3. Convert the operation string into the corresponding enumeration value.

4. Use a switch statement on the enumeration value.

5. Perform addition, subtraction or multiplication and print the result.

6. If the operation is invalid, display an error message.

C Program

```
#include <stdio.h>
#include <string.h>

enum Menu { ADD, SUBTRACT, MULTIPLY };

int main(void) {
    char operation[20];
    int a, b;
    enum Menu choice;

    scanf("%19s %d %d", operation, &a, &b);

    if (strcmp(operation, "ADD") == 0)
        choice = ADD;
    else if (strcmp(operation, "SUBTRACT") == 0)
        choice = SUBTRACT;
    else if (strcmp(operation, "MULTIPLY") == 0)
        choice = MULTIPLY;
    else {
        printf("Invalid choice\n");
        return 0;
    }

    switch (choice) {
        case ADD:
            printf("%d\n", a + b);
            break;
        case SUBTRACT:
            printf("%d\n", a - b);
            break;
        case MULTIPLY:
            printf("%d\n", a * b);
            break;
    }

    return 0;
}
```

Explanation

The enumeration gives meaningful names to the three menu choices. The entered operation is matched with its enum constant using strcmp(). The switch statement then selects the required arithmetic operation. For input ADD 10 20, the ADD case executes and prints 30.

Test Cases

Test Case	Input	Expected Output
1	ADD 10 20	30
2	SUBTRACT 20 8	12
3	MULTIPLY 6 7	42
4	DIVIDE 10 2	Invalid choice

Sample Output

```
Input:  ADD 10 20
Output: 30
```

Result

The program successfully uses an enumeration to represent menu choices and a switch statement to perform addition, subtraction and multiplication.